Pergamon

0956-7151(94)00285-1

Acta metall. mater. Vol. 43, No. 2, pp. 689–703, 1995
Elsevier Science Ltd. Printed in Great Britain

PREDICTION OF DENDRITIC GROWTH AND MICROSEGREGATION PATTERNS IN A BINARY ALLOY USING THE PHASE-FIELD METHOD

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(Received 22 April 1994)

Abstract—A comprehensive model is developed for solving the heat and solute diffusion equations during solidification that avoids tracking the liquid–solid interface. The bulk liquid and solid phases are treated as regular solutions and an order parameter (the phase field) is introduced to describe the interfacial region between them. Two-dimensional computations are performed for ideal solutions and for dendritic growth into an isothermal and highly supersaturated liquid phase. The dependence upon various material and computational parameters, including the approach to conventional sharp interface theories, is investigated. Realistic growth patterns are obtained that include the development, coarsening, and coalescence of secondary and tertiary dendrite arms. Microsegregation patterns are examined and compared for different values of the solid diffusion coefficient.

1. INTRODUCTION

The understanding and modeling of dendritic growth has remained a central theme of solidification research for decades. Significant progress has been made in understanding the transport of heat and solute in the vicinity of the dendrite tip as well as the factors that control the stability of the shape of the dendrite tip and determine the tip radius and initial secondary spacing. (See Glicksman and Marsh [1], Billia and Trivedi [2] for reviews of this subject.) To ultimately relate to the properties of castings, however, the major goal of dendritic modeling must be the prediction of chemical inhomogeneities (microsegregation) and their associated length scales that remain in the solid after solidification is complete. Thus various approaches have been made at modeling: (a) the transport of solute in the interdendritic mushy regions by diffusion [3, 4]; (b) the coarsening of the dendritic structure to reduce surface area [5–9]; and (c) the diffusion of solute in the solid [10–12]. A summary of much of this work is available in Kurz and Fisher [13]. The research in this paper presents a first attempt to model alloy dendritic growth with a single set of equations that includes all of these processes. As will be seen, however, this progress is not without its own difficulties in terms of computational requirements.

The central change in approach that permits this progress is the abandonment of the notion of a sharp transition between liquid and solid. Mathematically this sharp transition requires the tracking of the position and shape of the liquid–solid interface and the application of boundary conditions at a surface whose shape is part of the solution. By introducing a new variable, the phase field, that defines the physical state (liquid or solid) of the system at each point and its governing equation, one can employ numerical methods that avoid interface tracking. In contrast to enthalpy methods, which also avoid interface tracking, the introduction of a new variable that is defined independently of temperature and composition permits the existence of liquid regions that are undercooled or supersaturated and the inclusion of the effects of interface curvature. The transition between liquid and solid is represented by smooth but extremely localized changes in the phase field variable. The introduction of the phase field variable also permits the treatment of coalescence of solid particles and hence changes in the topology of the liquid–solid mixture.

From a physical point of view, this abandonment requires one to grapple with the physical nature of the liquid–solid interface. Whereas some structural information does exist for vapor–solid interface structure, little is known about the true structure of the interface between liquid and solid phases where the density is almost unchanged. Using Lennard–Jones potentials, molecular dynamics simulations of the transition in atomic positions across an interface have suggested that the interface width is several atomic dimensions [14]. Presently we will be unable to obtain usable simulations of dendritic growth with interface thickness in this range due to the limitations of computational resources.

Thus the interface width will be a parameter that affects the results of this paper. It should be realized, however, that in the limit as the interface thickness approaches zero the phase field equations converge to the sharp interface formulation [15, 16].

The basis of the phase-field model for solidification of pure materials goes back to efforts by Langer [17] and Caginalp [15], with related ideas in the context of anti-phase boundary migration (APB) by Allen and Cahn [18]. (In fact many of the ideas were elucidated by Cahn *et al.* in papers in 1960 [19] and 1964 [20].) Such models treat the presence of an interface by the introduction of a gradient correction to the thermodynamic potentials. Examples of application of this approach to dendritic solidification in pure materials can be found in Kobayashi [21] as well as Wheeler, Murray, and Schaefer (WMS) [22]. Because this approach solves a complete set of equations governing solidification, an *ad hoc* auxiliary condition specifying the operating tip radius is not required.

In the first section we present the formulation of the phase-field method for an alloy that involves both temperature and solute redistribution. Second we will pick a simplified lens-shaped phase diagram using ideal solutions to govern the bulk thermodynamics. Third we collect the equations necessary to treat the case where the solute alone controls the dendritic growth and treat the system as isothermal. This is called solution growth, and approximates the case for alloys containing large concentrations of alloying elements. Thus supersaturation will replace undercooling as the driving force for growth. We will show the relationship between the materials parameters of the phase-field method and the classical sharp interface formulation. Fourth we present two-dimensional calculations of dendritic growth at relatively large supersaturation (0.86) exploring the effects of mesh spacing, interface thickness, numerical noise, and ratio of liquid and solid diffusion coefficient. Finally we examine the issue of how thin the interface thickness needs to be in order to correspond to existing dendrite tip models.

2. DEVELOPMENT OF THE MODEL

The following is a synthesis of two phase-field formulations, an entropy formulation by Wang *et al.* [23] for a pure material and an isothermal energy formulation by Wheeler, Boettinger, and McFadden (WBM) [16] for a binary alloy. Similar models were also developed independently by Wheeler *et al.* [24], and Caginalp and Xie [25]. We begin our analysis in a manner analogous to Wang *et al.* by postulating the form for the entropy as

$$S = \int \left(s(\phi, e, c) - \frac{\epsilon^2}{2} |\nabla \phi|^2 \right) \mathrm{d}x^3 \qquad (2.1)$$

where s is the thermodynamic entropy density, ϕ is a phase field taking a value of zero in the solid and smoothly changing to one in the liquid, e is the internal energy density, and c is the concentration of solute B in solvent A. We require that $s(0, e, c)$ and $s(1, e, c)$ are the classical thermodynamic entropy densities for homogeneous solid and liquid. The parameter ϵ represents gradient corrections to the entropy density. Clearly, we have omitted the possibility of gradient corrections in c and e, the former having been investigated for solidification by Wheeler *et al.* [26]. We will also ignore volume changes, which should be a good approximation for the systems under consideration.

For dynamics, we may associate conserved fluxes with both the energy density e and the concentration c by writing

$$\dot{e} = -\nabla \cdot \mathbf{J}_e \qquad (2.2)$$

and

$$\dot{c} = -\nabla \cdot \mathbf{J}_c \qquad (2.3)$$

We are now ready to postulate forms for our fluxes in order to determine the equations of motion governing our system. Following the traditions of classical linear irreversible thermodynamics we assume that

$$\mathbf{J}_e = M_e \nabla \frac{\delta S}{\delta e} \qquad (2.4)$$

and

$$\mathbf{J}_c = M_c \nabla \frac{\delta S}{\delta c} \qquad (2.5)$$

where we have not allowed for off-diagonal terms which would allow for the Soret and Dufour effects. It is assumed that both M_c, and $M_e > 0$. Given that the phase field is not a conserved quantity, the simplest form for the evolution of the phase field is

$$\dot{\phi} = M_\phi \frac{\delta S}{\delta \phi}. \qquad (2.6)$$

Likewise we insist that $M_\phi > 0$. It is straightforward to show that equations (2.4)–(2.6) guarantee positive entropy production [27].

We now come to the first in a series of major approximations. For a full analysis of our system we should formally keep all the terms in the equations postulated above. Phenomenologically, we expect the effect of ∇c terms in the energy equation (2.2) and ∇T terms (where T is the temperature $T = \partial e / \partial s$) in the concentration equation (2.3) to be small corrections. We can achieve this simplification by insisting that

$$\dot{e} \approx -\left(\nabla \cdot M_e \nabla \frac{\delta S}{\delta e} \right)_{c \text{ fixed}} \qquad (2.7)$$

and

$$\dot{c} \approx -\left(\nabla \cdot M_c \nabla \frac{\delta S}{\delta c} \right)_{T \text{ fixed}}. \qquad (2.8)$$

These assumptions, coupled with equations (2.6), form the basis of this work, all other approximations

will concern only the relative values of various material parameters, including the phase diagram and the dependence of the free energy on the phase field.

In order that we may include the effects of anisotropy in the model, we follow the ideas put forth by [21], [22] and [28] and write

$$\epsilon = \bar{\epsilon}\eta = \bar{\epsilon}(1 + \gamma_\epsilon \cos(k_\epsilon \theta)) \tag{2.9}$$

where $\bar{\epsilon}$, γ_ϵ, and k_ϵ are constants, and

$$\tan\theta = \frac{\phi_y}{\phi_x} \tag{2.10}$$

and find that

$$\frac{\delta S}{\delta \phi} = \frac{\partial s}{\partial \phi} + \bar{\epsilon}^2 \nabla \cdot (\eta^2 \nabla \phi) - \bar{\epsilon}^2 \frac{\partial}{\partial x}\left(\eta\eta' \frac{\partial \phi}{\partial y}\right) + \bar{\epsilon}^2 \frac{\partial}{\partial y}\left(\eta\eta' \frac{\partial \phi}{\partial x}\right) \tag{2.11}$$

where $\eta' = \mathrm{d}\eta/\mathrm{d}\theta$. We also note that

$$\theta_x = \frac{\phi_x \phi_{xy} - \phi_y \phi_{xx}}{|\nabla\phi|^2}; \quad \theta_y = \frac{\phi_x \phi_{yy} - \phi_y \phi_{xy}}{|\nabla\phi|^2}. \tag{2.12}$$

Taking the variational derivative, and using basic thermodynamics, we find that

$$\frac{\delta S}{\delta c} = \frac{\partial s}{\partial c} = \frac{\mu^A - \mu^B}{T} \tag{2.13}$$

where μ^A and μ^B are the chemical potentials of components A and B and

$$\frac{\delta S}{\delta e} = \frac{\partial s}{\partial e} = \frac{1}{T}. \tag{2.14}$$

At this point, in order to make further progress, we need to specify how the chemical potentials depend on the phase field, temperature, and concentration. We introduce the Helmholtz free energy density f and write that

$$f = (1 - c)\mu^A + c\mu^B. \tag{2.15}$$

Here we assume that the chemical potentials μ^A and μ^B are those for a regular solution, and are written as

$$\mu^A = f^A(\phi, T) + \Omega(\phi)c^2 + \frac{RT}{v_m}\ln(1 - c) \tag{2.16}$$

and

$$\mu^B = f^B(\phi, T) + \Omega(\phi)(1 - c)^2 + \frac{RT}{v_m}\ln c. \tag{2.17}$$

Here $f^A(\phi, T)$ and $f^B(\phi, T)$ are the free energies of the pure materials, and each must have minima for all temperatures only at $\phi = 0$, and $\phi = 1$ (double well potentials). $\Omega(\phi)$ is a thermodynamic constant associated with the enthalpy of mixing, R is the gas constant, and v_m is the molar volume.

Now we must specify how f^A, f^B and Ω depend upon temperature and the phase field. This has already been done quite nicely for f^A and f^B in [23], and here we just state the results. Specifically, we assume that

$$e^A = e_S^A + p(\phi)(e_L^A - e_S^A) \tag{2.18}$$

where e_S^A and e_L^A are the solid and liquid internal energies of pure A, with analogous quantities being defined for $A \to B$. Here, we also assume

$$\Omega = \Omega_S + p(\phi)(\Omega_L - \Omega_S). \tag{2.19}$$

The parameters Ω_S and Ω_L are the enthalpies of mixing of the solid and liquid respectively. The factor $p(\phi)$ is a function which should be 0 in the solid and 1 in the liquid. Here it is chosen to have the form (after Wang *et al.* [23])

$$p(\phi) = \frac{\int_0^\phi g(\phi)\,\mathrm{d}\phi}{\int_0^1 g(\phi)\,\mathrm{d}\phi} = \phi^3(10 - 15\phi + 6\phi^2) \tag{2.20}$$

where we have taken

$$g(\phi) = \phi^2(1 - \phi)^2 \tag{2.21}$$

which is double-well potential with minima at $\phi = 0$, 1. The reasons for these particular choices for p and g are elucidated below. We will assume a constant heat capacity for the pure, single-phase materials, and hence the energies will be of the form

$$e_S^A(T) = e_S^A(T_m^A) + C_S^A(T - T_m^A),$$
$$e_L^A(T) = e_L^A(T_m^A) + C_L^A(T - T_m^A) \tag{2.22}$$

where T_m^A is the melting point of the pure material, and C_S^A and C_L^A are the heat capacities of the solid and liquid. For the materials of interest it is a good approximation to assume that

$$C_S^A = C_L^A = C^A. \tag{2.23}$$

It is also useful to define the latent heat

$$L^A = e_L^A(T_m^A) - e_S^A(T_m^A) \tag{2.24}$$

and finally write the energy as a function of phase field and temperature as

$$e^A = e_S^A(T_m^A) + C^A(T - T_m^A) + p(\phi)L^A. \tag{2.25}$$

Now using the thermodynamic relationship

$$f = e - Ts = e + T\frac{\partial f}{\partial T}$$

we find f^A to be

$$f^A = G^A(\phi)T + [e_S^A(T_m^A) - C^A T_m^A + p(\phi)L^A] \times \left(1 - \frac{T}{T_m^A}\right) - C^A T \ln\frac{T}{T_m^A} \tag{2.26}$$

where

$$G^A(\phi) = W^A g(\phi). \tag{2.27}$$

The function G^A is the double well $g(\phi)$, given in (2.21), scaled by the well-height W^A, a constant to be

determined below. We now see that the form of $p(\phi)$ is such that $p'(\phi) = 30g(\phi)$, which is zero for both $\phi = 0$ and $\phi = 1$. This insures that the minima of the free energy are at $\phi = 0$ (solid) and $\phi = 1$ (liquid), regardless of the temperature. The form of the free energy f^B is identical with all material parameters labeled by a superscript A replaced by $B(A \rightarrow B)$. As expected $f^A(0, T)$ and $f^A(1, T)$, the classical free energies of single phase solid and liquid respectively, are equal at the melting point T_m^A.

3. ADDITIONAL APPROXIMATIONS FOR COMPUTATIONAL SIMPLICITY

We now restrict our attention to a subset of the above formalism:

(1) We set $\Omega_L = \Omega_S = 0$, so that our solutions are ideal. This limits one to simple lens-type phase diagrams, but retains the essential feature of a "gap" in concentration between liquid and solid.

(2) We shall ignore the energy conservation equation (2.2) and instead impose a constant temperature field. This approximation will be valid as long as latent heat production is a small effect relative to the rejection of solute at the interface. Hence, for this work, we shall consider a purely isothermal model. Dendritic growth will be generated choosing a composition of liquid which is supersaturated with respect to the solid phase.

Using the above assumptions we can derive two simplified equations governing the evolution of our system. Our equation for $\dot{\phi}$ (2.6) can be found using

$$\frac{\partial s}{\partial \phi} = -\frac{1}{T}\frac{\partial f}{\partial \phi} = -(1-c)H^A(T,\phi) - cH^B(T,\phi) \tag{3.1}$$

to be

$$\begin{aligned}\dot{\phi}/M_\phi = {} & [\bar{\epsilon}^2\eta^2\nabla^2\phi - (1-c)H^A - cH^B] \\ & + \bar{\epsilon}^2\eta\eta'[\sin(2\theta)(\phi_{yy} - \phi_{xx}) \\ & + 2\cos(2\theta)\phi_{xy}] - (1/2)\bar{\epsilon}^2[\eta'^2 + \eta\eta''] \\ & \times [2\sin(2\theta)\phi_{xy} - \nabla^2\phi - \cos(2\theta)(\phi_{yy} - \phi_{xx})].\end{aligned} \tag{3.2}$$

Here H^A and H^B are defined by

$$\begin{aligned}H^A(T,\phi) &= \frac{1}{T}\frac{\partial f^A}{\partial \phi} = W^A g' + 30gL^A\left(\frac{1}{T} - \frac{1}{T_m^A}\right) \\ H^B(T,\phi) &= \frac{1}{T}\frac{\partial f^B}{\partial \phi} = W^B g' + 30gL^B\left(\frac{1}{T} - \frac{1}{T_m^B}\right).\end{aligned} \tag{3.3}$$

We may also allow M_ϕ to depend on composition, and so postulate a form

$$M_\phi = (1-c)M^A + cM^B. \tag{3.4}$$

Similarly, we may find the solute conservation equation (2.3) by noting that

$$\frac{1}{T}\nabla(\mu_A - \mu_B) = (H^A - H^B)\nabla\phi - \frac{R}{v_m}\frac{\nabla c}{c(1-c)} \tag{3.5}$$

and holding T fixed we find

$$\dot{c} = \nabla \cdot M_c\left[\frac{R}{v_m}\frac{\nabla c}{c(1-c)} + (H^B - H^A)\nabla\phi\right]. \tag{3.6}$$

In the same manner that we let M_ϕ depend upon c we shall let M_c depend upon ϕ and postulate

$$M_c = \frac{v_m}{R}c(1-c)[D_S - p(\phi)(D_L - D_S)] \tag{3.7}$$

where D_S and D_L are the classical diffusion coefficients in the solid and liquid. The $c(1-c)$ factor in equation (3.7) has been chosen so the diffusion coefficients in the solid and liquid are constants. This allows us to write equation (3.6) in the slightly more familiar looking form

$$\dot{c} = \nabla \cdot D_c\left[\nabla c + \frac{v_m}{R}c(1-c)(H^B - H^A)\nabla\phi\right] \tag{3.8}$$

with

$$D_c = D_S + p(\phi)(D_L - D_S). \tag{3.9}$$

One notes that for a single phase liquid or solid, $\phi = 0$ or 1 everywhere, and equation (3.8) is simply the ordinary solute diffusion equation. Equations (3.2) and (3.8) are the equations governing the evolution of the system used in the remainder of this paper.

4. MATERIAL PARAMETERS

Just as we related M_c to the measurable quantities D_L and D_S, we must now relate the remaining parameters in our theory to other experimentally realizable measurements. It is unnecessary to go through the details of this subject as it has been done well elsewhere using an energy formulation [16]. Here we write down the relevant parameters, which can be checked dimensionally.

Using ideas developed by Allen and Cahn [18] and others we may write the surface energy for a pure material as

$$\sigma^A = \epsilon\sqrt{2}\int_0^1 \sqrt{T_m^A f^A(T_m^A, \phi)}\, d\phi. \tag{4.1}$$

This determines W^A as

$$W^A = 18\left(\frac{\sigma^A}{T_m^A \epsilon}\right)^2. \tag{4.2}$$

The thickness of the interface δ^A is defined (analogously to WBM) by solving the one-dimensional phase-field equation for isothermal coexistence of liquid and solid at a planar interface of pure A to find

$$\phi(x) = \frac{1}{2}\left[1 + \tanh\left[\frac{\sqrt{W^A}x}{\sqrt{2}\epsilon}\right]\right] \tag{4.3}$$

where x is the spatial coordinate, and then identifying

$$\delta^A = \frac{\epsilon}{\sqrt{4W^A}}. \tag{4.4}$$

Note that with this definition ϕ changes from 0.1 to 0.9 over a distance of approx. $6\delta^A$. M^A is given by

$$M^A\epsilon^2 = \frac{\sigma^A T_m^A \beta^A}{L^A}, \tag{4.5}$$

where β^A is the linear kinetic coefficient for interface attachment that relates interface undercooling to interface speed by $V_n = \beta^A(T_m^A - T)$. Since all of the above equations can also be written for A→B, and ϵ is assumed to be independent of concentration, there is a restriction on the independence of our material parameters, specifically

$$\frac{\sigma^A\delta^A}{T_m^A} = \frac{\sigma^B\delta^B}{T_m^B}. \tag{4.6}$$

This condition is not particularly stringent, because, for small enough δ^A and δ^B, our results should be independent of δ^A and δ^B.

Although M^A, M^B, ϵ, W^A, and W^B appear naturally in our expressions for the phase-field dynamics, they are not parameters which are typically employed in solidification. With this in mind we rewrite the above relations so that, given σ^A, δ^A, δ^B, β^A and β^B, we can determine these less intuitive parameters. We find

$$M^A = \frac{(T_m^A)^2\beta^A}{6\sqrt{2}L^A\delta^A}; \quad M^B = \frac{(T_m^B)^2\beta^B}{6\sqrt{2}L^B\delta^B};$$
$$W^A = \frac{3\sigma^A}{\sqrt{2}T_m^A\delta^A}; \quad W^B = \frac{3\sigma^B}{\sqrt{2}T_m^B\delta^B};$$
$$\epsilon^2 = \frac{6\sqrt{2}\sigma^A\delta^A}{T_m^A} = \frac{6\sqrt{2}\sigma^B\delta^B}{T_m^B}. \tag{4.7}$$

5. NUMERICAL TECHNIQUES AND OTHER COMPUTATIONAL ISSUES

5.1. Finite difference techniques

One of the major advantages of phase-field models is that they remove the necessity of tracking the interface. In particular, the model can be solved with boundary conditions only applied at the edges of a computational box. This quality also allows very simple-minded computation schemes to yield interesting results.

For our simulations we solve equations (3.2) and (3.8) on a two-dimensional uniform grid using second order in space, and first order in time-finite difference approximations to the derivatives. We call the grid spacing Δx and the time step Δt. We should note that both equations are in fact diffusion equations with diffusivities $D = D_c$ or $D = M_\phi\epsilon^2$. Conservative differencing [29] was used for the concentration equation so that mass conservation was ensured.

The choice of the Δt must be made in such a way that the equations remain stable under time-step iteration. For the diffusion equation in two dimensions it can be shown that

$$\Delta t < \frac{\Delta x^2}{4D} \tag{5.1}$$

for stability, while the presence of a non-linear source term in all likelihood would require smaller time steps than this restriction. Given that we chose $M\epsilon^2 \approx D_L$ for reasons to be discussed in the next section, we found, by numerical experimentation, that a value of $\Delta t = (\Delta x)^2/5D_L$ works well for not too large choices of the anisotropy.

5.2. The interface thickness

Upon differencing our equations we must then decide upon a reasonable value for the interface thickness δ. If we had infinite computational resources then we might choose the interface thickness to be extremely small (perhaps 1 nm). This choice, however, requires that the *entire* grid be resolved at that scale, as we have used such a simple finite difference technique. If we examine equation (3.2) with $\delta^A = \delta^B = \delta$ the difficulties become apparent. We have

$$\dot\phi = \frac{M_\phi\epsilon^2}{\delta^2}\left[\delta^2\nabla^2\phi - \frac{1}{4}\frac{\partial g}{\partial\phi} - \frac{5g\delta}{\sqrt{2}d_0^A} \times\left[(1-c)\left(\frac{T_m^A}{T}-1\right) + c\frac{d_0^A}{d_0^B}\left(\frac{T_m^B}{T}-1\right)\right]\right] \tag{5.2}$$

where we have used the capillary length $d_0^A = \sigma^A/L^A$. Braun *et al.* [30] have shown that, in order to properly recover sharp interface morphological stability results, we must make $\delta \leqslant d_0$. Unfortunately, for a metal $d_0 \approx 5 \times 10^{-8}$ cm, so if we would like to resolve micron length structures would require many mesh points indeed. In the next section we will investigate what the practical limits are towards resolving large scale microstructure, which will in turn impose limits on our value of δ.

5.3. Noise

One matter which we have not explored yet is the inclusion of stochastic noise. We need such noise to stimulate fluctuations at the interface, which, when amplified, give rise to much of the structure observed in real systems. Here we have chosen to include the noise in a rather *ad hoc*, yet simple manner. Specifically, if we examine equation (3.2), we see that noise can be introduced by

$$\dot\phi/M_\phi \to \dot\phi/M_\phi - \alpha r(16g(\phi))[(1-c)H^A + cH^B] \tag{5.3}$$

where r is a random number between $+1$ and -1, and α gives the magnitude of the fluctuation. Note that the factor $16g(\phi)$ has a maximum value of 1 at $\phi = 0.5$ and drops off rapidly away from the interface. It should be noted that a more rigorous method

of introducing noise—a full Langevin treatment—is possible [31], but was not deemed necessary for our model.

Using the above scheme a code was developed in the C Programming Language, and implemented on a Cray Y-MP4E/232, and achieved full vectorization of the differential equations. Speeds on the order of 150–180 MFLOPS were typical.

6. RESULTS FOR SPECIFIC MATERIAL PARAMETERS

As we elucidated above, we would very much like to use real material parameters in our simulations. Unfortunately, the two goals of obtaining realistic looking microstructure and extremely thin interfaces stand in computational opposition. To examine the asymptotic limit towards sharp interface formulations, it is necessary to let δ^A and δ^B get as small as is computationally feasible, while to observe large-scale microstructures requires large computational "boxes". In order to explore these issues it was found helpful to think about the problem as a theoretical progression from large scale microstructure, computed with a thick interface, to a single, barely developed tip computed with a thin interface.

For concreteness we use materials parameters appropriate to nickel–copper, because this system exhibits a nearly ideal solution behavior [16]. We take (where A is nickel and B is copper) $T_m^A = 1728$ K, $T_m^B = 1358$ K, $L^A = 2350$ J/cm^3, $L^B = 1728$ J/cm^3, $\sigma^A = 3.7 \times 10^{-5}$ J/cm^2, $\sigma^B = 2.9 \times 10^{-5}$ J/cm^2, and $D_L^A = D_L^B = 10^{-5}$ cm^2/s. For simplification we have set v_m equal to 7.42 cm^3 for both elements.

The only other material parameters that we need to specify are the interface kinetic coefficients and the solid diffusion constant. For practical reasons we found it necessary to significantly reduce the values of β^A and β^B from the estimates made by Coriell and Turnbull [32], which apply to collision limited growth. Instead we employ values that would apply when the rate at which atoms join the crystal is controlled by diffusive jumps. Thus both the phase field and liquid concentration fields will have the same "diffusivities". There is no reason why a more sophisticated computational technique, or a far more time consuming set of computations, could not be done with the larger values. Hence we chose $\beta^A = 0.33$ cm/K/s and $\beta^B = 0.39$ cm/K/s. This means that $M_\phi \epsilon^2 \approx D_L$.

For a typical metal alloy such as nickel–copper the diffusion is substitutional, and $D_S \approx 10^{-9}$ cm^2/s near the melting point, while for an alloy governed by interstitial diffusion this constant can be 1000 times larger so that $D_S \approx 10^{-6}$ cm^2/s. We will keep our diffusivity at the smaller value for much of our analysis, and use an interstitial value to examine the effect of different D_S upon solute segregation patterns.

Having chosen our materials parameters, we must now specify interface thickness, the supersaturation, and the size of computation box, so that we can observe realistic dendritic microstructures. This involves a rather simple, but necessary, analytical argument using some of the more basic results of classical dendrite theory.

The argument presented here is a classic example of needing to know the answer to the question before one can answer the question. This type of dilemma exists in many numerical, as well as laboratory, experiments. The starting points for our analysis is the classical Ivantsov solution to the dendrite problem. This formula provides a relationship between the supersaturation and the Peclet number $p = \rho v / 2D_L$, where ρ is the tip radius and v is the tip velocity. The relationship in two dimensions is

$$\Delta = \frac{c_L - c_\infty}{c_L - c_S} = Iv(p) = \sqrt{\pi p} e^p \operatorname{erfc}(\sqrt{p}) \quad (6.1)$$

where c_L and c_S are the concentrations at the dendrite tip in the liquid and solid, respectively, and c_∞ is the concentration far ahead of the interface. Equation (6.1) is correct only for $\sigma^A = \sigma^B = 0$ and $\beta^A = \beta^B = \infty$.

The other formula needed is the condition of "marginal stability" which, while hardly rigorous, provides a good guess at the proper tip radius [33]. We have

$$\sigma^* = \frac{l_0}{(c_L - c_S) p \rho} \quad (6.2)$$

where $l_0 = \sigma^A T / L^A |dT/dc|$ is another capillary length closely related to, but not the same as, d_0 appearing in 5.2, and where $|dT/dc|$ is the liquidus slope. For models with no diffusion in the solid we expect $\sigma^* \approx 0.05$ [34].

Our hope is to determine a supersaturation such that ρ, which will necessarily be larger than l_0, will be sufficiently smaller than the dimensions of the box. For ideal asymptotics we would like δ to be as small as possible ($\delta \approx l_0$), but for the moment it is better to find any tip, and then improve our resolution.

In what follows we shall always assume that $\delta \approx \Delta x$, and make determinations with that assumption in mind. It turns out that we can get an idea of the minimum size of our grid by recognizing two important restrictions on Δx. First we need to resolve the tip adequately, hence

$$\Delta x < \frac{\rho}{10} \quad (6.3)$$

where clearly the divisor 10 is some approximate value. Second we need the dendrite to fit nicely in the box, and grow a good distance, so we require that

$$N \Delta x = L > 50 \rho \quad (6.4)$$

where N is the number of grid points in one linear

dimension, L is the length of the box in one dimension, and 50 is again an approximate value. Combining the above two expressions we easily find that

$$N > 500. \tag{6.5}$$

This restriction is of course only approximate, but is a good estimate. It is interesting that it can be computed regardless of the materials parameters governing the system.

We may also make a determination of the Peclet number necessary to perform the calculation in a "reasonable" amount of time. Again we have two restrictions, this time on Δt. First, in order to guarantee numerical stability we require

$$\Delta t < \frac{\Delta x^2}{5D_L} \tag{6.6}$$

as we discussed in the previous section. Second, we want the dendrite to transverse the entire box. Letting $N = 500$, and calling M the number of time steps we then have that

$$M\Delta t > \frac{L}{v} \tag{6.7}$$

where v is the velocity of the dendrite. Combining 6.6 and 6.7 yields

$$p > \frac{12500}{M}. \tag{6.8}$$

Clearly, to limit computation time, we would like to minimize the number of time steps, hence we may compute dendritic patterns most easily at high Peclet number $p \approx 1$.

The problem is not even this simple, however, even if we wanted to run a calculation with $p \approx 0.1$, and M ten times larger we encounter an additional problem. Since $p\rho$ is approximately fixed for a given temperature by equation (6.2) if p is decreased by a factor of 10 than ρ must be increased by the same factor. Based on the above arguments, in order to view an entire dendrite we must increase the box size by either increasing Δx, which then brings us further from the asymptotic regime, or increase the number of mesh points, making the computation more difficult. Hence, large Peclet number is also the only domain where we have a good chance of getting acceptable asymptotics in a reasonable amount of time.

In the limit of large Peclet number we can expand 6.1 to read

$$\Delta \approx 1 - \frac{1}{2p} \tag{6.9}$$

which can then be combined with 6.2 to yield

$$\frac{\rho}{l_0} \approx 40\frac{1-\Delta}{(c_L - c_S)} = 40\frac{c_S - c_\infty}{(c_L - c_S)^2} \tag{6.10}$$

where we have substituted $\sigma^* = 0.05$. It is quite clear that one needs to carefully choose the supersaturation to guarantee that ρ not be too large. The values chosen were $T = 1574$, yielding equilibrium flat interface concentrations of $c_L^{eq} = 0.4668$ and $c_S^{eq} = 0.3994$, and $c_\infty = 0.40831$, yielding an initial supersaturation of $\Delta = 0.86$.

After manipulating the number of mesh points N, the resolution (Δx), choosing an anisotropy of $\gamma_4 = 0.04$ with four-fold symmetry, and a noise amplitude of $\alpha = 0.3$, we found an acceptable looking dendritic microstructure, which appears in Fig. 1. Although complete histories of the development of dendritic structure have been computed, we have chosen to show only one time slice for each run. In this diagram the concentration profile is given everywhere, while in the lower half $\phi = 0.1 \rightarrow 0.9$ contours are shown in black. Thus one can gain a measure of the diffuseness of the interface compared to the microstructural detail. The initial condition was a small diamond of solid ($\phi = 0$) in the center of a liquid ($\phi = 1$), with the concentration everywhere equal to c_∞. The calculation was in fact only done in one quadrant, $N = 750$, and then reflected about the x and y axes, where Neumann boundary conditions were imposed. Thus a pair of artificial two-fold symmetry lines occur. In full simulations with growth from the center of a computational domain, the growth exhibits a general four-fold symmetry, but the details of the side branching are, of course, not perfectly symmetric. It should be noted that the solution of the phase-field equations leads naturally to selection of a tip radius and growth velocity without extra conditions. Coalescence has occurred between some sidebranches and the solid contains microsegregation patterns that will be discussed in detail below. The pictured box size is 6.9×10^{-3} cm, $N = 1500$, and $0.94(\delta) = \Delta x = 4.6 \times 10^{-6}$ cm. In this figure, the ratio of $\delta/l_0 \approx 68$, which is quite large ($l_0 \approx 7.1 \times 10^{-8}$ cm for our calculation). The tip speed was approx. 1 cm/s, and the tip radius was approx. 5.1×10^{-5} cm. Note that in this, and in all subsequent calculations we have chosen to set $\delta^A = \delta^B = \delta$. We cannot hope to obtain accurate asymptotics from this run, but the features are qualitatively interesting, and the deviation in the maximum value of c_L from asymptotics is only $\approx 7\%$. We will address the issues surrounding asymptotics later. The run took approx. 7500 CPU seconds on the Cray Y-MP4E/232.

6.1. *Solute segregation patterns and microstructure*

Before embarking upon a detailed discussion of microstructure, and resultant solute patterns, we would like to examine the effects of computational parameters Δx, δ and α. In Fig. 2(a) we have a calculation with all the same parameters as in Fig. 1, except that, while holding Δx fixed we have reduced the number of mesh points, $N = 500$, thereby reducing the physical dimensions of our box. Here we have chosen not to fold the plot over. We will compare all other plots to Fig. 2(a).

We would like to be sure that as we increase the

resolution (Δx decreased) that the patterns are not significantly affected. Figure 2(b) shows the same calculation as Fig. 2(a) with 1.5 times the resolution ($\Delta x \rightarrow \Delta x/1.5$), and similar patterns do indeed develop.

The variation in δ is a straightforward matter, and we have shown in Fig. 2(c) the result of increasing δ by 1.5. As is not too surprising, the likelihood of coalescence increases with increases in δ. Also, we note the tip radius has increased, and the tip speed decreased. This is also not surprising, because, as we increase δ, small wavelength perturbations will be swamped out by the phase field, and hence longer modes will be enhanced. In general, we found that it was desirable to keep δ as small as possible, and still have $\delta \geqslant \Delta x$, where the equality holds in Figs 1 and 2(a). We must recall that good asymptotics are expected as δ decreases, and as we are far from this limit, it only worsens the quality of the results to make δ larger than it needs to be.

The only other concern to be addressed might be the behavior and effect of our noise term. All the previous figures shown are for fluctuations of the source term with $\alpha = 0.3$. In Fig. 2(d)–(f) we show the same set of parameters as in Fig. 2(a) but now with the maximum amplitude of α fixed at 0.075, 0.01875 and 0 respectively. The first feature of note is that the sidebranches do indeed disappear when the noise is turned off, indicating that numerical noise (due to the finite difference approximation and roundoff) is not the source of sidebranches for this set of parameters. Also since the sidebranches develop nearer the tip for higher noise, they therefore grow to a longer length in the same amount of time.

Having examined the effects of various computational parameters, we now turn to a more detailed discussion of the observed microstructure in Fig. 1. The calculated pattern shows many features consistent with experience based on observations of dendritic growth in transparent materials and with metallographic examination of cast microstructures. First the spine of the primary stalk has a low concentration, and the regions of the dendrite between the secondary arms have the highest concentration. Second, the tip–tip competition of the sidebranches tends to favor the amplification of every other sidebranch. This is consistent with the predictions of Warren *et al.* [35] in a different setting, viz. dynamic primary wavelength selection during directional solidification. Third, we note the tendency of the sidebranches to begin their growth in a direction not perpendicular to the primary stalk, and to later develop a growth axis that is perpendicular to the primary stalk. Behind the dendrite tip the root of the side arm is seen to be significantly narrower than further out the secondary arm. This feature is commonly observed in real dendrites. A marked asymmetry of growth is noted: the side of the secondary arm towards the front of the dendrite thickens more. It is interesting to note the existence of spines of low concentration on secondary branches that are as low as the spines in the primary stalks. The asymmetry of the shape of the side arms with respect to this spine is quite noticeable. Fourth, we note the coalescence of adjacent and impinging secondary and tertiary arms. (The black line denoting contours of the phase field can be distinguished from the color of the solute pattern.) This coalescence process surely occurs in dendritic structure where little or no eutectic forms as is the case for the lens type phase diagram used here. In the areas of coalescence, a solute content is left in the solidified neck region that is intermediate between the compositions of the solid and liquid regions before coalescence. To the authors' knowledge, this has not been noted in metallographic examinations and is probaby due to the extremely high local solidification rate that occurs to eliminate negative curvature just after initial contact. (The details of coalescence are sensitive to the value of δ used for the calculation.)

We may also examine solute patterns for the case where the solid diffusivity is much larger. In Fig. 3, a comparison of two values of D_S/D_L, 10^{-4} [Fig. 3(a), (c), (e)] and 10^{-1} [Fig. 3(b), (d), (f)] is shown. An increased level of solid diffusion is known to reduce the severity of microsegregation and this is shown nicely in the calculated patterns. By comparing the composition contour line for 0.399 near the center of the primary stalks in Fig. 3(a) [same as Fig. 2(a)] and (b), we see the effect of this increase. This contour disappears some distance behind the tip in Fig. 3(b) but remains in (a) all the way back to the seed. Thus the extremely low composition region at the center of the primary stalk is eliminated.

In Fig. 3(c) and (d) the microsegregation pattern near the tip is shown at higher magnification with three contours levels of solute content. One notes that smaller lateral gradients of solute exist in the solid at the tip as well as behind the tip when the solid diffusion is faster. Interestingly, the tip radius appears to depend slightly on the level of solid diffusion. The increased diffusion in the solid decreases the tip radius about 4% and also decreases the growth velocity. We are unsure of the general validity of this observation, but note that Barbieri and Langer [34] have predicted an increase of the product $\rho^2 v$ when thermal diffusion is present in the solid for dendritic growth into a pure undercooled melt. Clearly this topic requires closer examination.

Figure 3(e) and (f) show enlargements from the interdendritic regions behind the tips for the two cases. Here the liquid regions are blue. In the solid, high solute gradients as well as green areas associated with coalescence are observed when the solid diffusion is limited. The green areas have disappeared in Fig. 3(f). One notes the variation in solid composition (orange to yellow) and liquid composition (medium blue to dark blue) associated with the positive and negative curvatures respectively of the

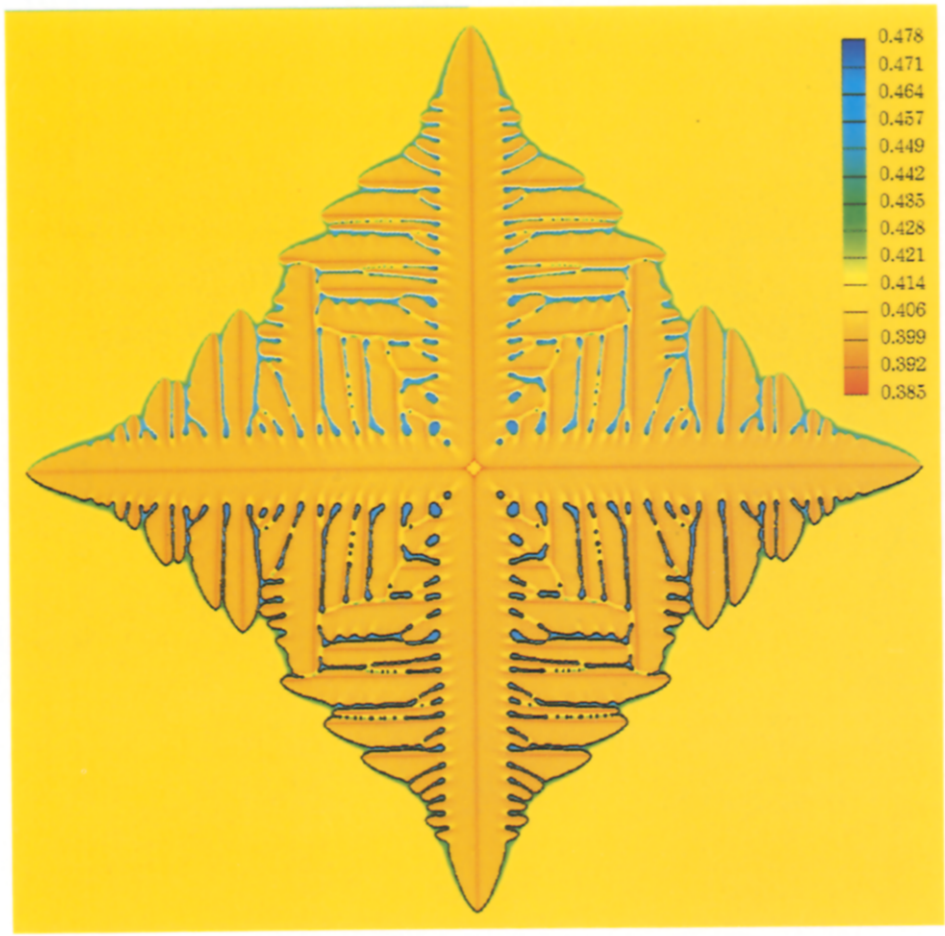


Fig. 1. A plot of the concentration and phase field profiles for a nickel–copper alloy. The calculation was done on a 750 × 750 grid, and then reflected about the x and y axes. The colors range from red (lowest concentration) to green (central concentrations) to blue (highest concentrations). All the color figures employ this color scheme. The dark black lines shown in the lower half of the figure are the phase field contours from $\phi = 0.1$ to 0.9. In this plot the diffusion rate in the solid is 10^{-4} times smaller than the liquid diffusion rate.

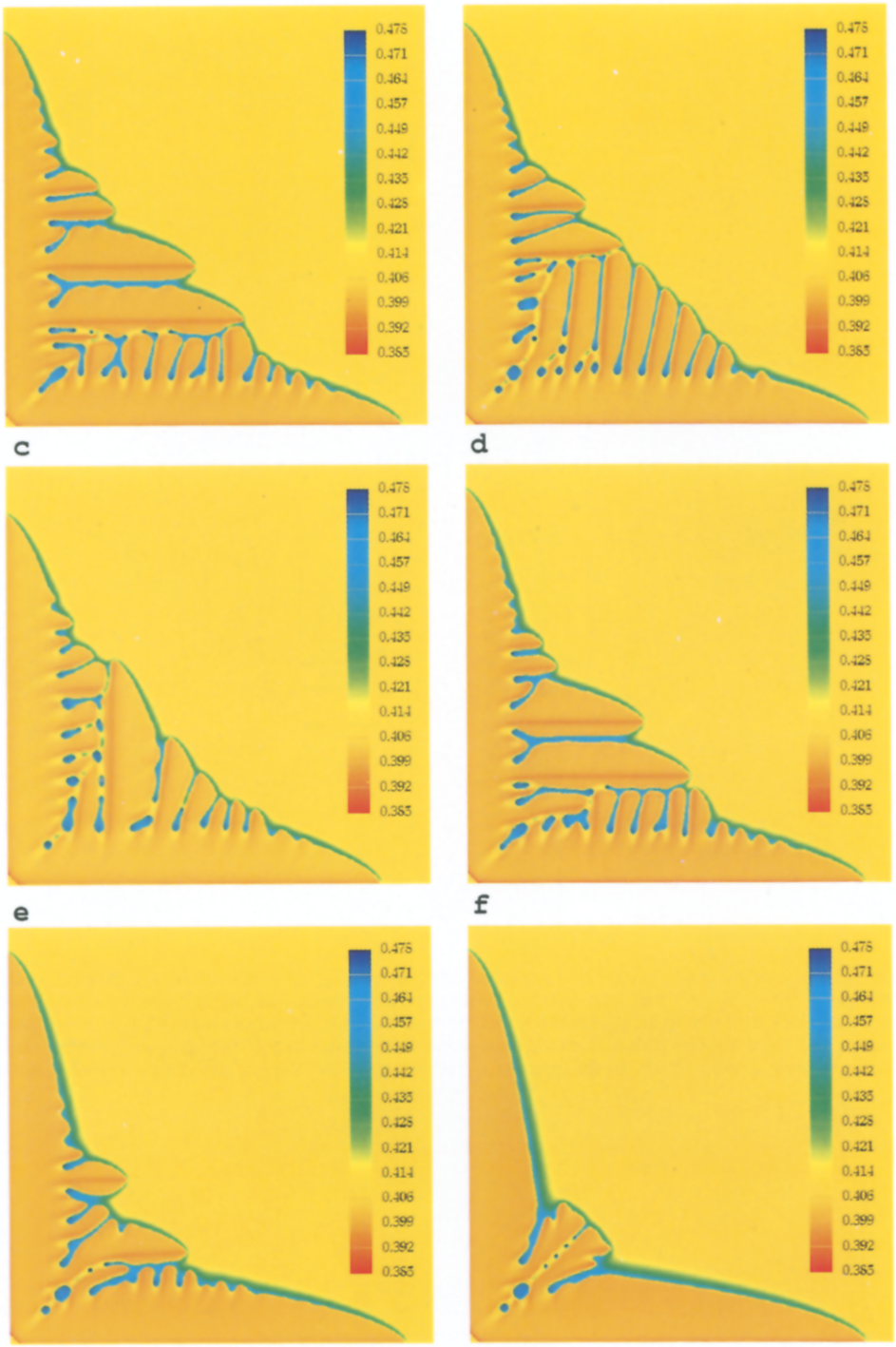


Fig. 2. Plots of concentration profiles obtained from calculations using a computational domain 2/3 the size of one quadrant in Fig. 1. In (a) all other parameters are the same as in Fig. 1 and in (b)–(f) only one other parameter was varied: (b) the value of Δx was decreased by 2/3; (c) the interface thickness was increased by 3/2; (d)–(f) the noise amplitude α was reduced to 0.075, 0.01875 and 0.

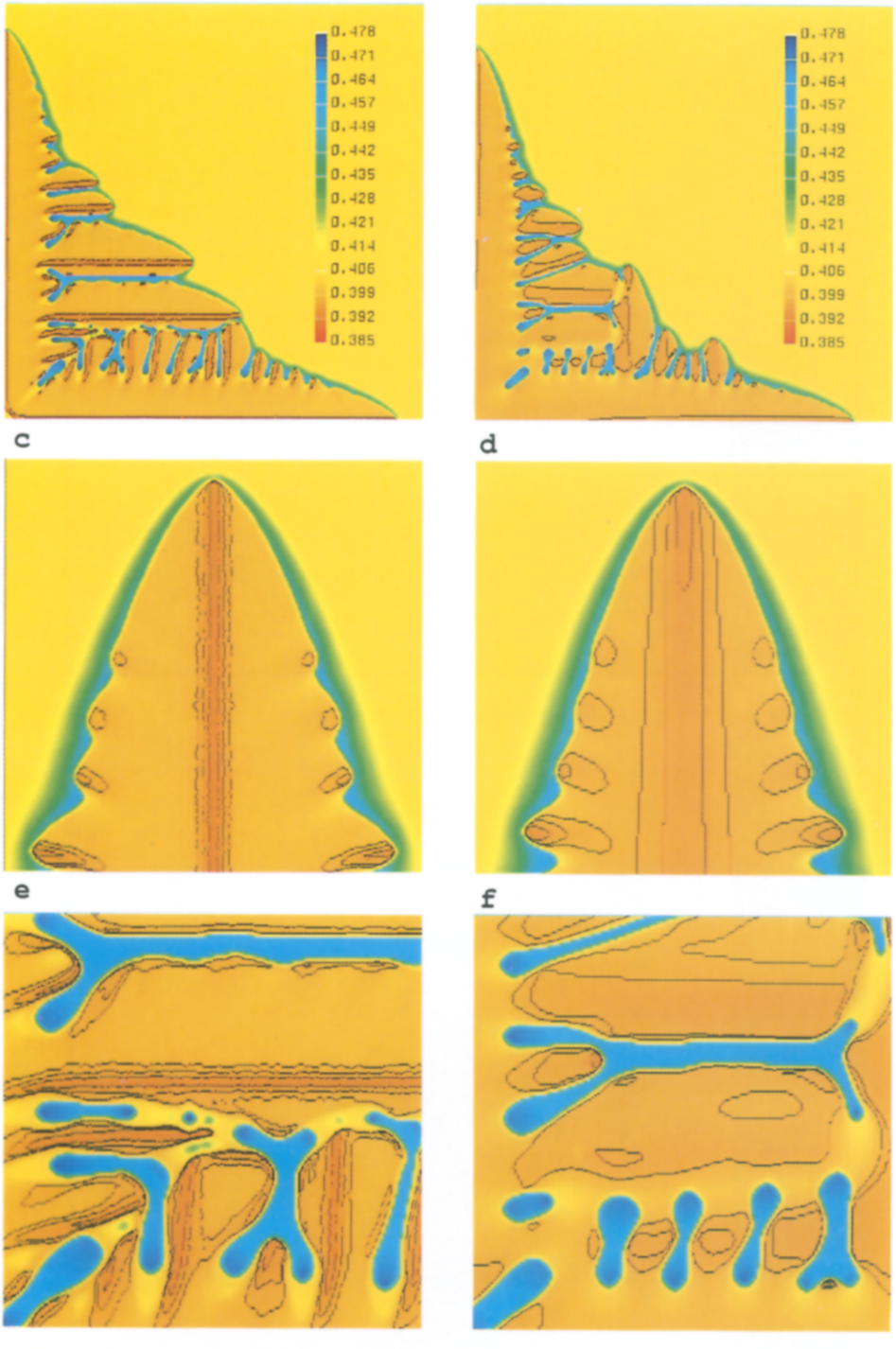


Fig. 3. Plots of the concentration profile, showing the effects of variations in the solid diffusivity. (a), (c) and (e) show the solute profile for $D_S/D_L = 10^{-4}$, and (b), (d) and (e) show the solute profile for $D_S/D_L = 10^{-1}$. (a) and (b) show the entire pattern, (c) and (d) show the tip region and (e) and (f) show an interdendritic region. In (a) and (b) we have included, as a black line, the solute contour $c = 0.399$, while in (c)–(f) we have included three contours for $c = 0.397$, 0.399 and 0.4005.

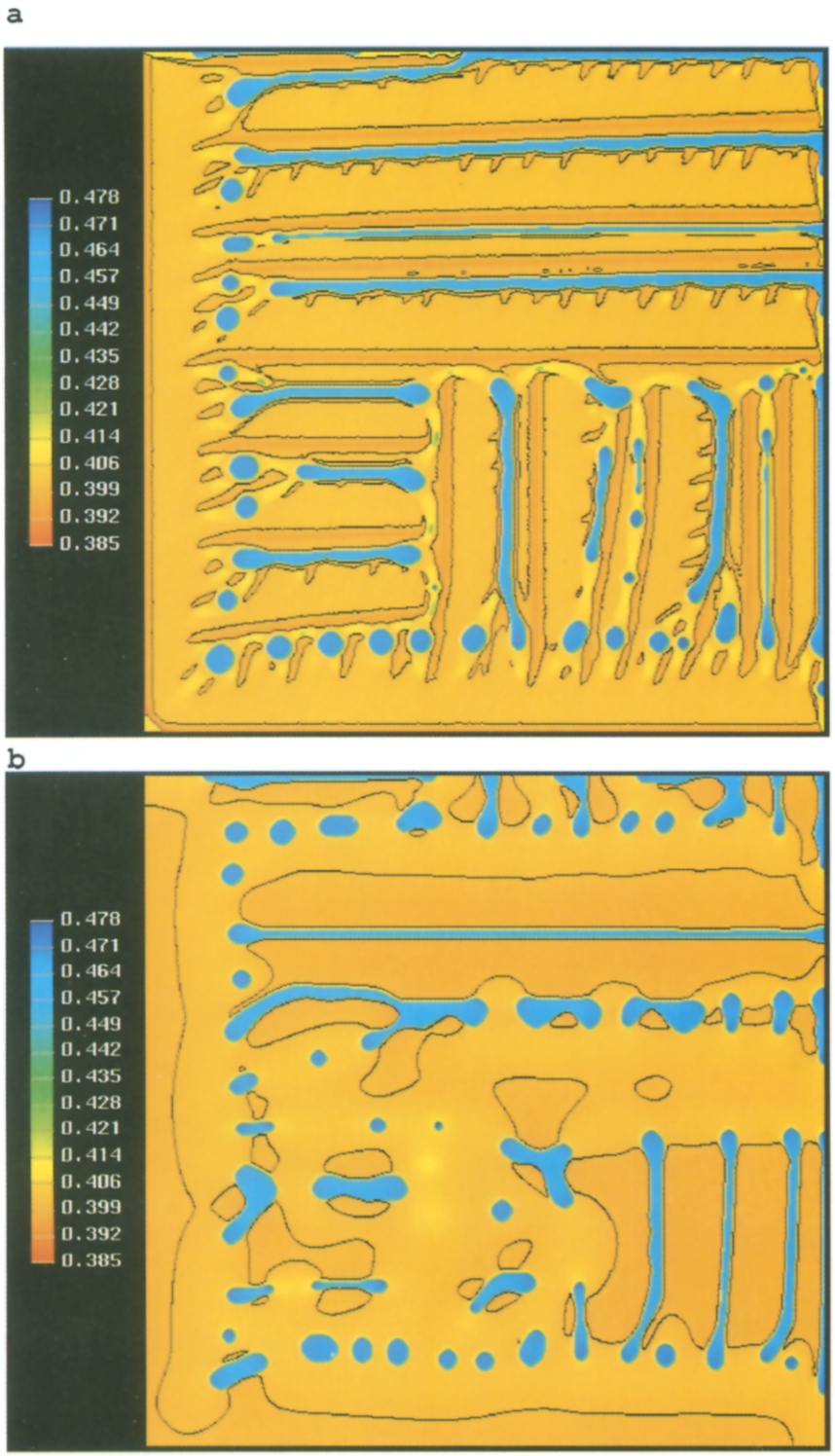


Fig. 4. Plots of the concentration profile, where the computations used for Fig. 3(a) and (b) have been run three times as long, and hence the dendrite impinges with the walls

liquid–solid interface. Long term coarsening is driven by these composition differences.

The computations shown in Fig. 3(a) and (b) were allowed to run for additional time until the dendrites collided with the edge of the computational box as shown in Fig. 4(a) and (b). Because there is a no-flux condition at the boundary, this might be thought to simulate the impingement of equiaxed dendritic grains that originate from separate nucleation events. The area fraction solidified is approximately equal to $\Delta = 0.86$.

It is informative to describe the solute segregation patterns that develop during dendritic solidification in a more quantitative way. Typically, solidified microstructure are characterized experimentally by metallographic sectioning and characterization using X-ray microprobe analysis. Line scans that traverse the distance from the center of a dendrite arm out to the last region to solidify are normally employed. A few authors [36, 37] have used area scans or randomly selected line scans where a histogram of fraction of measurements occurring in various concentration bins can be constructed. They presented this data as concentration vs fraction of measurements with equal or lesser concentration (the integral of the frequency histogram), a procedure that yields a plot of composition vs fraction solidified. Such plots are compared to simple models of solute segregation such as the Scheil equation or modification thereof for solid (back) diffusion. We can plot the analogous histograms to examine our computed segregation patterns.

Figure 5(a)–(c) show segregation histograms corresponding to Figs 3(a), 4(a) and 4(b) respectively. In Fig. 5(a), three major peaks with increasing levels of concentration are seen corresponding to solid, extradendritic liquid, and interdendritic liquid. The solid peak is relatively broad, containing information regarding the microsegregation pattern in the frozen solid. It contains concentrations below and above the equilibrium solid composition for a stationary, flat interface. The concentration in the solid at the dendrite tip is the lowest observed in the solid. Its value is the combined effect of the shift in the solid composition due to curvature (and kinetics). This composition is discussed in Section 6.2 and compared to results of the sharp interface model of dendritic growth. The central peak is quite sharp and corresponds to the bulk of the liquid which still remains at c_∞. We note a tail on the high concentration side corresponding to the build up of solute around the dendrite envelope. The small peak at the highest concentration corresponds to the interdendritic liquid between the dendrite side branches.

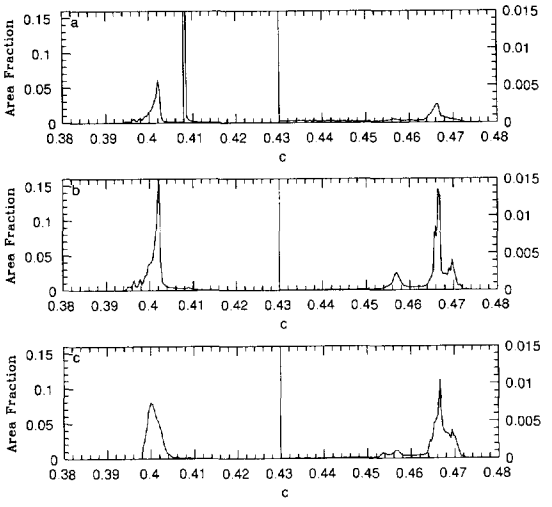


Fig. 5. Histograms (a)–(c) of the fraction of area vs concentrations for Figs 3(a), 4(a), and 4(b) respectively. An expanded vertical scale is used to the right of the vertical line in the center of each histogram. The equilibrium values for a stationary flat interface are 0.3994 and 0.4668 solid and liquid respectively. The initial concentration is $c_\infty = 0.40831$.

In Fig. 5(b) the central peak of Fig. 5(a) is gone. The peak at low solute concentration corresponds to the solid. The interdendritic liquid peak is seen to contain a small spread in composition reflecting the positive and negative curvatures of the interface between the liquid and solid. A small intermediate peak at 0.41 corresponds to the regions of solid where coalescence has occurred as described previously. Figure 5(c) shows the effect of increased solid diffusion on the segregation histogram. We note the disappearance of the lowest concentration spikes, corresponding to the elimination of the central core of depleted solute in the dendrite stalks and side-branches. The solid concentration peak observed in Fig. 5(b) has shifted and decreased in amplitude in Fig. 5(c). The range in concentrations seen in Fig. 5(c) reflect primarily the variations in interface curvature. Long term coarsening of this structure would lead to sharpening of this peak at the solid concentration given by the phase diagram.

Despite the realism of calculated pattern the reader should be cautioned against overinterpretation of the results. For solidification of Ni–Cu alloy, latent heat released during growth into a supersaturated liquid cannot be neglected. The consequential temperature rise during growth could significantly alter the solute pattern.

6.2. *Asymptotics*

It is important to compare our results with the results of sharp-interface theory. To do this, it is necessary to reduce δ such that it approaches l_0. This is no simple matter, unfortunately, as grids which are dense enough are difficult to achieve for boxes that still contain a entire tip, as we argued above. It was found feasible to reduce δ down to $4l_0$, on a box that just barely contained a tip. This box was 13 times smaller than the box given in Fig. 1. We should note that this computation took 72,000 CPUs on the Cray Y-MP4E/232. To get the resolution truly needed would require another order of magnitude in computer speed, just to obtain sharp interface results for

Table 1

δ/l_0	c_L/c_L^{SI}	$\Delta_{mod}/\Delta(p)$	ρ/l_0
68.0	0.93	0.58	733
23.7	0.95	0.74	409
6.4	0.97	0.87	216
3.1	0.98	0.91	206

In the above δ is the interface thickness, l_0 is the capillary length, c_L is the liquid concentration at the interface and Δ_{mod} is the interface supersaturation, both obtained from the simulations. C_L^{SI} is the sharp interface prediction of equations (6.11) and (6.12), and $\Delta(p)$ is the Ivantsov prediction both computed for values of tip radius and velocity obtained from the simulations.

a single tip, let alone for a large microstructure. Here we must be content with showing convergence towards the asymptotic limit, and not exact correspondence.

In the asymptotic limit of a sharp interface, WBM [16] have shown that the isothermal alloy phase-field equations predict that the compositions on either side of the liquid–solid interface are related by the parallel tangent construction. The energy difference between the parallel tangents to the liquid and solid free energies is related to the normal velocity and the curvature. For our present model of ideal solutions (with $\delta^A = \delta^B$ and $M^A = M^B$), this condition is given by

$$\left(\frac{v_n}{\beta^A \eta(\theta)} + \frac{\mathcal{K}\sigma^A(\eta(\theta)+\eta''(\theta))T_m^A}{L^A}\right) = \left(\frac{(T_m^A)^2}{T} - T_m^A\right) + \frac{R(T_m^A)^2}{v_m L^A}\ln\frac{1-c_L}{1-c_S} \quad (6.11)$$

and

$$\left(\frac{v_n}{\beta^B \eta(\theta)} + \frac{\mathcal{K}\sigma^B(\eta(\theta)+\eta''(\theta))T_m^B}{L^B}\right) = \left(\frac{(T_m^B)^2}{T} - T_m^B\right) + \frac{R(T_m^B)^2}{v_m L^B}\ln\frac{c_L}{c_S} \quad (6.12)$$

where v_n is the velocity of the interface in the outward normal direction (into the liquid), $\mathcal{K}$ is the curvature of the interface, and we are reminded that $\eta(\theta) = 1 + \gamma_\epsilon \cos(4\theta)$. Note that the changing compositions around the contour of the dendrite interface can be understood by a consideration of these parallel tangents and their separation based on local velocity and curvature (including signs). The $A \rightarrow B$ symmetry of equations (6.11) and (6.12) is a consequence of equation (4.6).

If β^A and β^B were infinite, expressions 6.11 and 6.12 reproduce precisely the Gibbs–Thomson effect normally employed as interface conditions in sharp interface models where local equilibrium is assumed. As shown in [26], inclusion of a gradient energy term for the concentration breaks the requirement for parallel tangent in the asymptotic limit and leads to solute trapping. However, this term was not included in the present paper.

As a check on our numerical calculations and the entrie formulation, comparisons were made to these asymptotic expressions of (6.11) and (6.12) and to the Ivantsov model in Table 1. The value of the liquid composition at the interface was extracted from the diffuse interface calculations as the maximum value of composition in the interfacial region.

As we can see the convergence towards sharp interface theories improves with decreasing δ, as the ratio of $c_L/c_L^{SI} \rightarrow 1$, but is not yet so good that we can use such data to accurately predict interface concentrations to within 1%.

We may also compare our data with the predictions of the Ivantsov model, given by equation (6.1). Here we can compare the predicted value of the supersaturation Δ, given p from the simulation, and directly measure Δ at the tip, the so called modified Ivantsov solution, given by

$$\Delta_{mod} = \frac{c_L^{tip} - c_\infty}{c_L^{tip} - c_S^{tip}}. \quad (6.13)$$

Again we see convergence towards expected values, as given in Table 1. Perhaps most telling is the dependence of ρ on the value of δ. As we mentioned earlier, as we increase δ we will suppress short wavelength modes that might be expressed, hence causing ρ to be larger than the sharp interface theories would predict. Table 1 shows the dependence of ρ on δ, and it does indeed decrease with decreasing δ.

7. CONCLUSIONS

We have developed a robust model of binary alloy solidification, which we have used to simulate the isothermal growth of large scale dendritic microstructures into a highly supersaturated liquid, and to predict the pattern of solute distribution within the solid and liquid. The research presented in this paper is a first attempt to model alloy dendritic growth with a single set of evolutionary equations. Realistic growth patterns were obtained that include the development, coarsening, and coalescence of secondary and tertiary dendrite arms. While such predictions are still relatively far from the asymptotic limit, we have confidence that the results are of the correct general character.

We see several areas as requiring more theoretical attention. First, we would like to include the effects of heat flow in our calculation, instead of limiting ourselves to the isothermal calculation presented here. This is not a trivial modification from a computational viewpoint, since the thermal field responds on a much faster time scale than the solute field, but there may be approximation schemes which overcome this difficulty.

Second, we would like our calculations to be closer to the asymptotic limit. When we are able to perform calculations with $\delta \rightarrow d_0$, we will then be able to make a quantitative comparison of our results with laboratory experiments, instead of the qualitative comparison used here. One should note that a proper test of microscopic solvability theories will demand a

very thin interface. Indeed, as a computer science problem, the solution of these equations was effected in the most conceptually simple matter, which in all likelihood is far from the most efficient method. Using such methods we must wait perhaps 10 yr for computers to reach the power needed to do a large scale, asymptotically correct simulations, hence we must investigate the possibility of more sophisticated algorithms. In particular, it would seem that adaptive grid methods, which would only require small Δx near the interface, are a most promising candidate for tackling this system of equations.

Last, we hope that our techniques will provide insight into a variety of other solidification phenomena including simulations of directional growth (constant temperature gradient), coarsening [38], inclusion of ∇e and ∇c terms in the entropy density S, which would allow for new surface energy effects and solute trapping in the asymptotic limit [26], nucleation theory, and the study of tip–tip competition in order to better understand the primary spacing problem.

Acknowledgements—We would like to particularly thank B. T. Murray for his insights and suggestions. We would also like to thank R. J. Braun, G. B. McFadden, R. F. Sekerka, A. R. Roosen, and A. A. Wheeler for helpful discussions and advice.

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